VENDRA - Production Deployment Changes

Backend on Render, Frontend on Vercel, Google login preserved.

1. The main cause (why it would have broken)

In development the Vite dev server proxied every **/api** call to localhost:3000, so the browser saw one origin. On Vercel there is no Vite proxy, so **/api/...** would hit Vercel itself and return 404. On top of that: CORS was commented out, the Google login redirects were hard-coded to **http://localhost:5173**, the Google callback URL was relative (it would resolve to the Render host, so the login cookie would be set on the wrong domain), cookies had no production flags, and refreshing a page such as /cart on Vercel would 404 because there was no SPA fallback.

2. The approach: Vercel proxies /api to Render

A rewrite in **Frontend/vercel.json** forwards **/api/*** to the Render service. The browser only ever talks to the Vercel domain, so (a) cookies are first-party and are not blocked as third-party cookies (Safari/Chrome), (b) no CORS problems, (c) the frontend code (axios baseURL '/api/...', Google button '/api/auth/google') needs no change. The Google redirect URI is therefore the **Vercel** URL, so the login cookie is set on the Vercel domain.

3. File-by-file changes

File	Change	Reason
Frontend/vercel.json (new)	Rewrite /api/:path* to Render; all other paths to /index.html	Replaces the Vite proxy; fixes 404 on refresh of React Router pages
Backend/src/config/config.js	Added FRONTEND_URL, GOOGLE_CALLBACK_URL, REDIS_TLS; both URLs are required when NODE_ENV=production	Removes hard-coded localhost; fails fast if misconfigured
Backend/src/app.js	Enabled CORS (origin FRONTEND_URL, credentials); trust proxy; /health route; Google callbackURL from env with proxy:true	Safe behind Render's proxy; health check for Render; correct Google redirect URI
Backend/src/controllers/user.controller.js	Cookie options: httpOnly, secure in production, sameSite lax, 7-day maxAge; Google redirect goes to FRONTEND_URL; clearCookie uses same options; added missing return statements in register/login (status for duplicate user now 409)	Secure cookies over HTTPS; logout actually clears cookie; prevents 'headers already sent' crash / null user crash
Backend/src/routes/auth.route.js	Google failureRedirect uses FRONTEND_URL/login	Was localhost or a relative path
Backend/src/dao/cache.js	Optional TLS when REDIS_TLS=true	Managed Redis (Upstash etc.) usually requires TLS
Backend/package.json	start script, engines node >=20	Render runs 'npm start'
Backend/.env.example (new)	Lists all required env vars	Reference for Render dashboard

4. Render setup (Backend)

Root Directory: **Backend** | Build Command: **npm install** | Start Command: **npm start** | Health Check Path: **/health**

Environment variables: NODE_ENV=production, MONGO_URI, JWT_SECRET, GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, FRONTEND_URL, GOOGLE_CALLBACK_URL, IMAGEKIT_PRIVATE_KEY, RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET, REDIS_HOST, REDIS_PORT, REDIS_PASSWORD, REDIS_TLS. Do not set PORT; Render provides it.

MongoDB Atlas: Network Access must allow Render (0.0.0.0/0 on the free plan).

5. Vercel setup (Frontend)

Root Directory: **Frontend** | Framework: Vite | Build: npm run build | Output: dist

Environment variable: **VITE_RAZORPAY_KEY_ID** (Razorpay key id). Do NOT set a base URL variable; the app uses /api via the rewrite.

6. Google Cloud Console (needed for Google login)

Credentials > your OAuth client:
Authorized JavaScript origins: **https://YOUR-APP.vercel.app**
Authorized redirect URIs: **https://YOUR-APP.vercel.app/api/auth/google/callback**
The same URL must be the GOOGLE_CALLBACK_URL value on Render. Keep the localhost entries for local development.

7. Values still required

Replace **YOUR-RENDER-SERVICE.onrender.com** in Frontend/vercel.json with the real Render URL, and set FRONTEND_URL / GOOGLE_CALLBACK_URL with the real Vercel URL. Deploy order: Render first (get URL) -> put URL in vercel.json -> deploy Vercel (get URL) -> set Render env vars and Google Console.

8. Known notes

Render free tier sleeps after inactivity (first request is slow). If the Google 'Authorized redirect URI' mismatches you will see redirect_uri_mismatch. Local development is unchanged (defaults fall back to localhost and the Vite proxy).